

Work Stream 2: Deriving the Thermal Energy Equation and Vaporization Limit

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1 Introduction

A central question raised by [u/vibe0009](#) in [r/FluidMechanics](#) was how enstrophy divergence implies thermal shock when the incompressible Navier-Stokes equations lack a temperature equation. To address this, we must step outside the idealized incompressible frame and couple the viscous dissipation to the thermal energy equation.

2 Thermal Coupling

The complete energy equation for a fluid includes the viscous dissipation function Φ :

$$\rho c_p \left(\frac{\partial T}{\partial t} + u \cdot \nabla T \right) = k \nabla^2 T + 2\mu \mathbf{D}(u) : \mathbf{D}(u) \quad (1)$$

The contraction of the strain rate tensor $\mathbf{D}(u)$ is directly proportional to the enstrophy density $|\nabla \times u|^2$ in incompressible flow.

3 Vaporization by Pure Force

As [u/Little-Name9809](#) pointed out, this suggests a mechanism for vaporizing a fluid into plasma using only mechanical force (and no external heat). Because the OpenAI singularity requires the local enstrophy density to scale as $\tau^{-2.515}$, the local heat generation rate diverges.

Integrating this rate against the thermal diffusion timescale reveals that the fluid reaches 373 K (boiling) at $\tau \approx 45.6$ ps prior to the mathematical blowup. The continuum assumption is shredded by the phase change.

4 Acknowledgments

Thank you to [u/vibe0009](#) and [u/Little-Name9809](#) for prompting this explicit thermal derivation. *Rendons à César ce qui appartient à César.*